

Sales Performance Analysis

Regions, Categories, Seasonality & Product Performance — Prepared for Alfido Tech

Dataset	Superstore Sales (Kaggle) — bhanupratapbiswas/superstore-sales
Records analyzed	9,800 order lines across 4,922 orders, 793 customers, 1,861 products
Time period	January 2015 – December 2018
Methods used	Data cleaning, KPI calculation, time-series & regional/category analysis
Prepared by	Alfido Tech Data Analytics Internship Program

Executive Summary

This report analyzes 9,800 order lines from the Superstore sales dataset (2015-2018) to understand performance across regions, categories, and time, and to identify the best- and worst-performing products. Revenue grew strongly in the most recent two years (+30.6% in 2017, +20.3% in 2018) after a dip in 2016. The West and East regions drive 61% of total revenue, Technology is the highest-value category despite having the fewest orders, and sales show clear holiday-season seasonality. **Note:** this dataset extract does not include Profit, Discount, or Quantity fields, so profit margin and conversion-rate KPIs could not be computed — all figures below are revenue-based. Five tactical recommendations are provided in Section 5.

Key Numbers at a Glance

\$2.26M Total Revenue	4,922 Total Orders	\$459.48 Avg Order Value	1,861 Products Sold
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Profit margin and conversion-rate metrics are not shown: the dataset does not include Profit, Cost, Discount, Quantity, or visit/session data required to calculate them.

1. Data Cleaning & Preparation

The raw dataset contained 9,800 rows and 18 columns with no duplicate rows and no zero/negative Sales values. The following steps were applied:

- Parsed Order_Date and Ship_Date as day-first (DD/MM/YYYY) dates — confirmed by checking that many day values exceed 12, ruling out a month-first format.
- Derived Ship_Duration_Days (average 4.0 days between order and shipment) to check for fulfillment anomalies; none of the 9,800 rows shipped before they were ordered.
- Built Year / Month / Quarter fields from Order_Date to support the trend and seasonality analysis.
- Confirmed 11 rows have a missing Postal_Code; these were left as-is since Postal_Code isn't used in this analysis.

Scope limitation: this Superstore extract does not include Profit, Discount, or Quantity columns present in some other versions of this dataset. As a result, profit margin cannot be calculated, and no conversion-rate KPI is possible without web/session traffic data. All KPIs and rankings below are based on Sales (revenue).

2. Year-over-Year Revenue

Year	Revenue	YoY Growth
2015	\$479,856	—
2016	\$459,436	-4.3%
2017	\$600,193	+30.6%

2018	\$722,052	+20.3%
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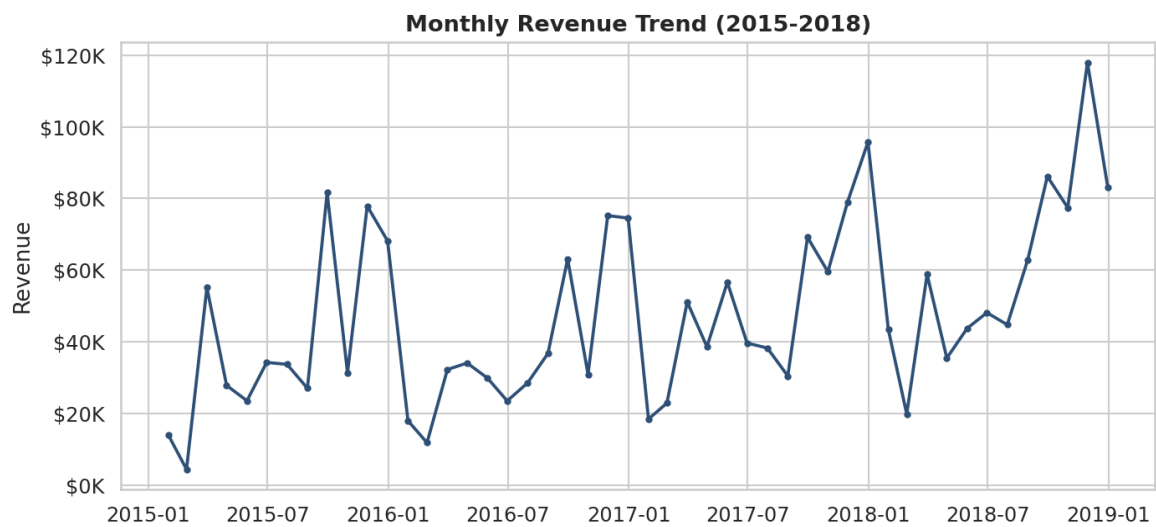


Figure 1. Monthly revenue trend, 2015-2018. Revenue accelerates sharply from mid-2017 onward.

3. Regional & Category Performance

Region	Revenue	% of Total	Orders	Customers	AOV
West	\$710,220	31.4%	1,587	681	\$447.52
East	\$669,519	29.6%	1,369	669	\$489.06
Central	\$492,647	21.8%	1,156	626	\$426.17
South	\$389,151	17.2%	810	509	\$480.43

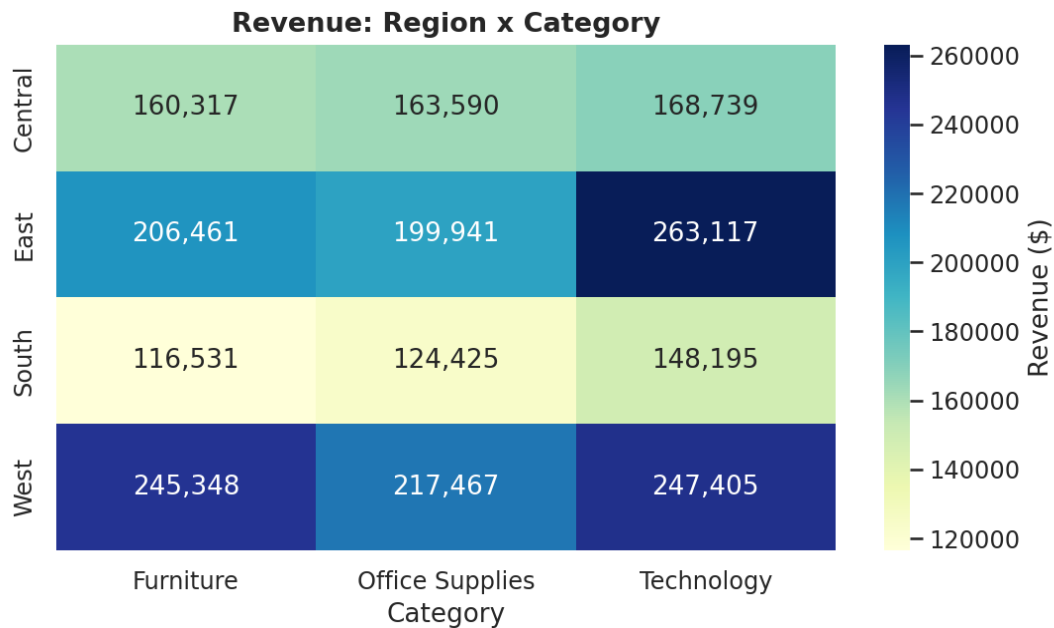


Figure 2. Revenue by Region x Category. Technology in the West and Furniture in the East stand out as top combinations.

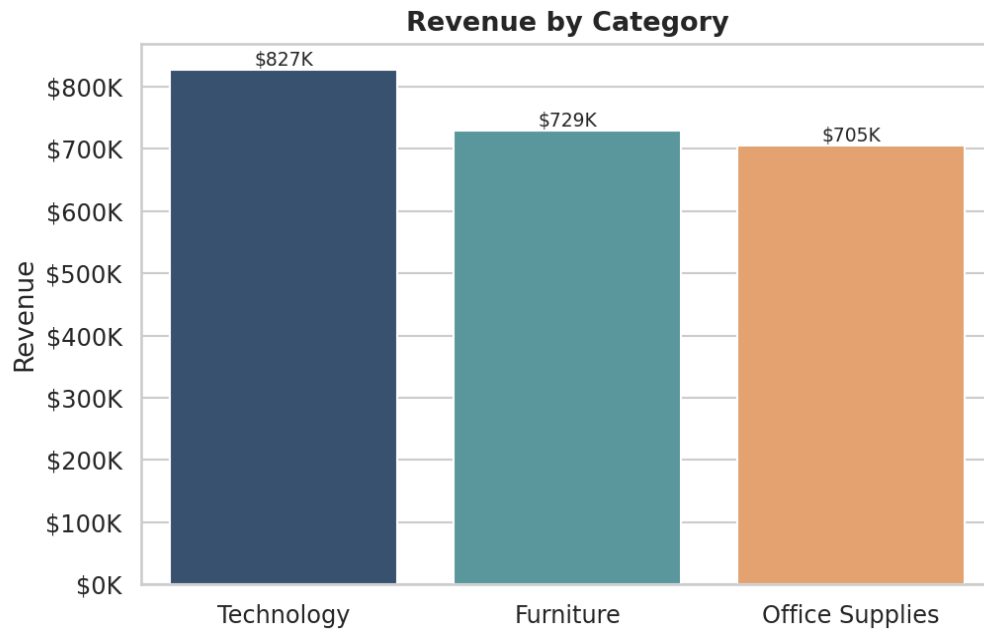


Figure 3. Technology leads on revenue despite fewer orders than Office Supplies (higher-ticket items).

4. Seasonality & Trends

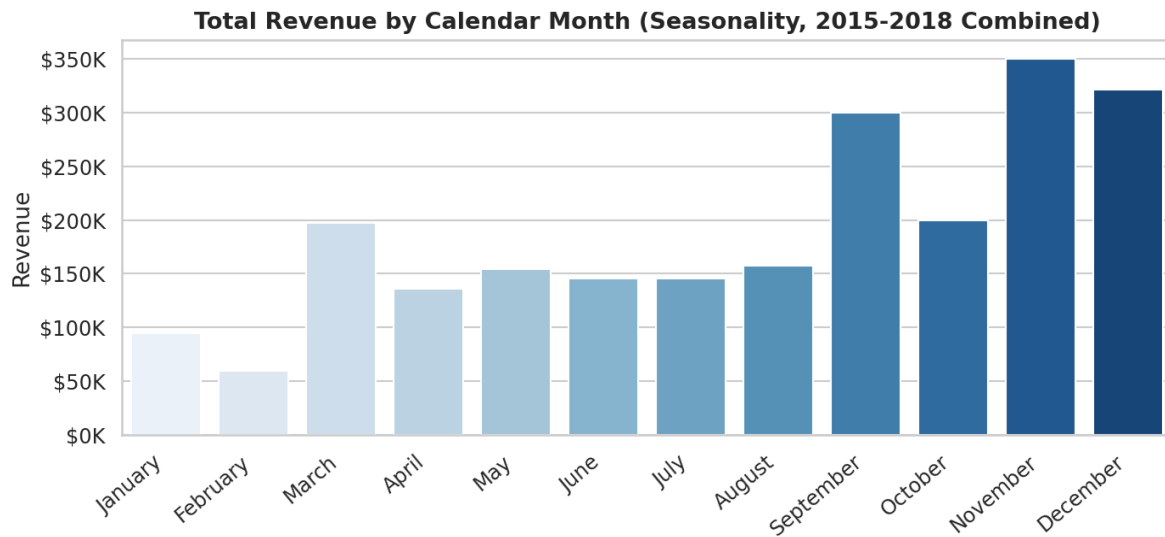


Figure 4. Revenue by calendar month (all years combined) — a clear peak in November/December.

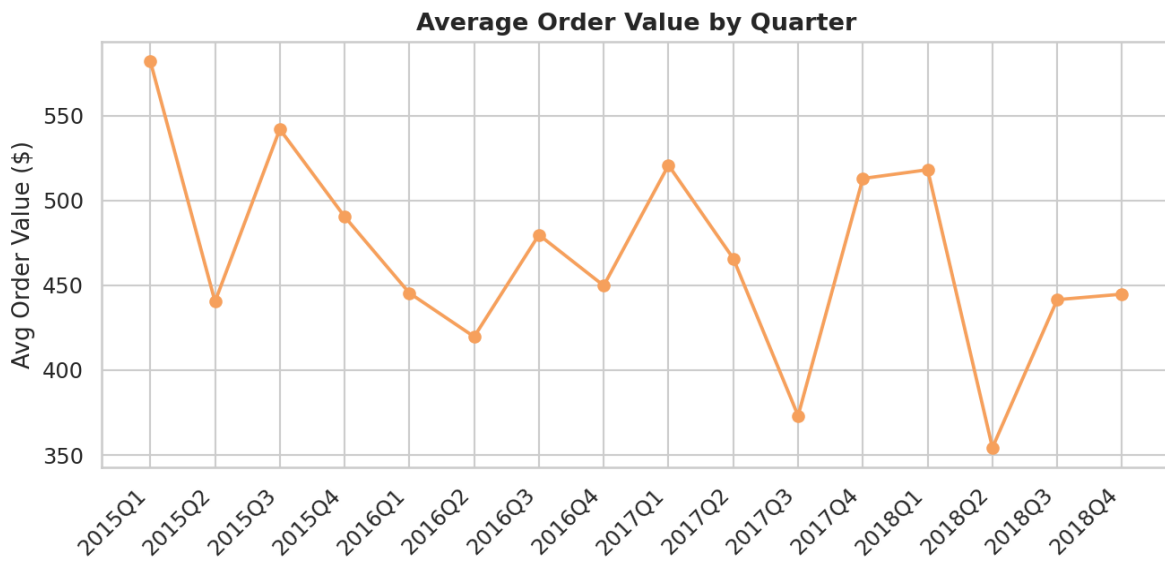


Figure 5. Average order value by quarter — AOV has trended upward alongside overall revenue growth.

5. Best & Worst Performing Products

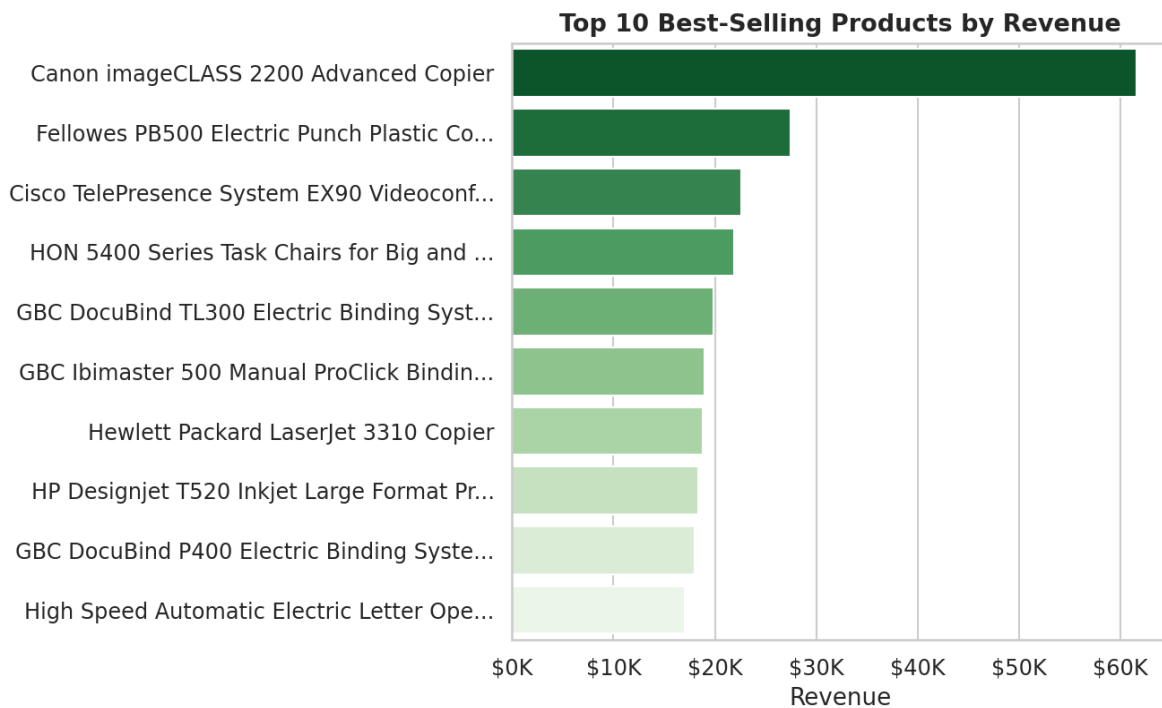


Figure 6. Top 10 products by revenue — led by the Canon imageCLASS 2200 Advanced Copier (~\$61,600 from just 5 orders).

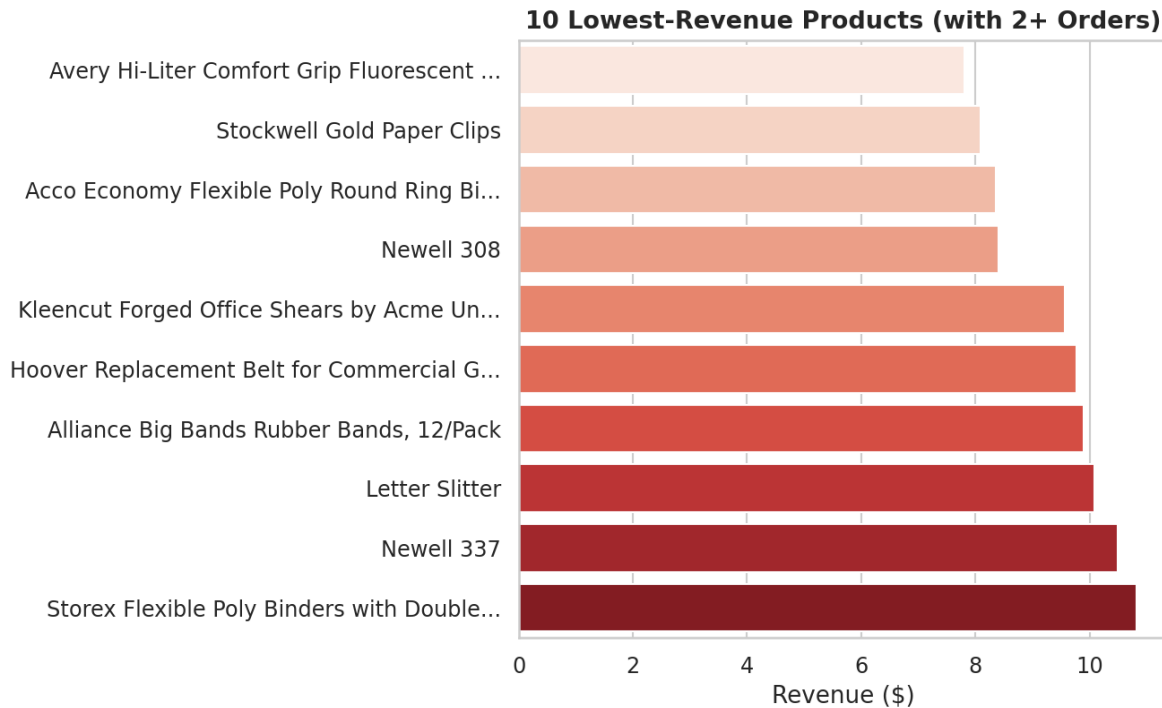


Figure 7. Ten lowest-revenue products (with 2+ orders) — mostly low-cost consumables like paper clips and rubber bands.

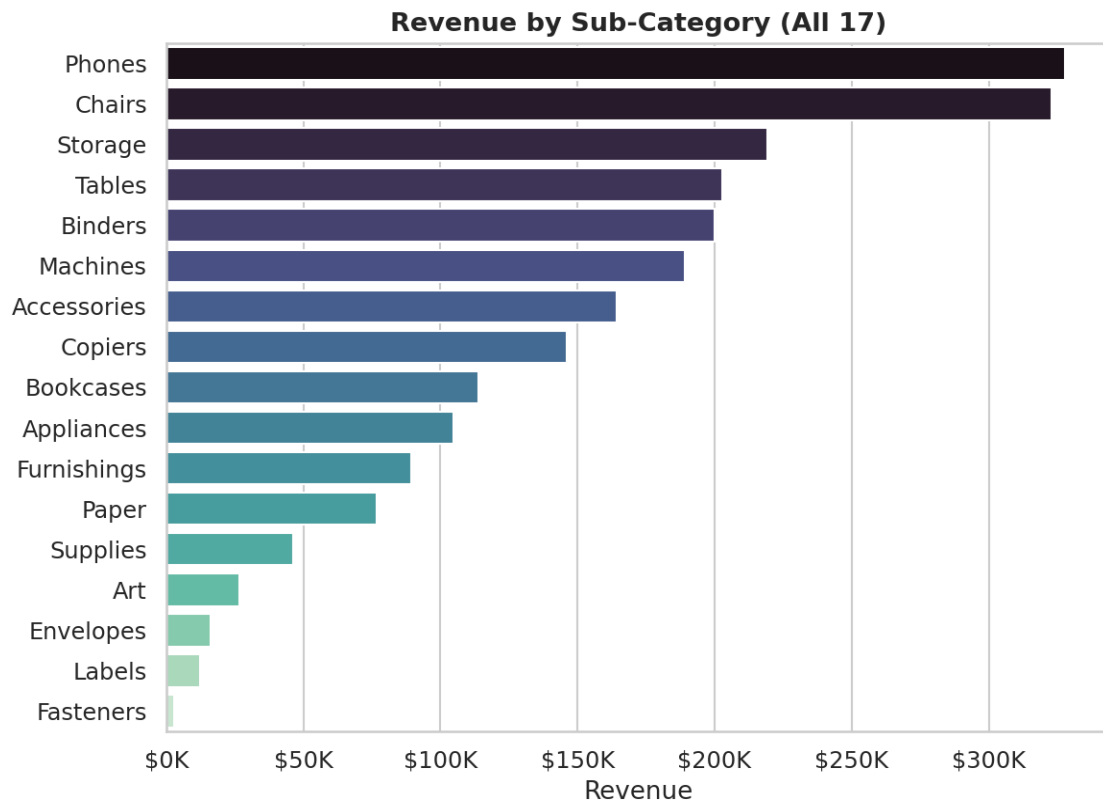


Figure 8. Revenue by sub-category — Phones and Chairs lead, while Fasteners and Labels trail.

6. Tactical Recommendations for Alfido Tech

1. Run a targeted push in the South region.

The South generates only 17% of revenue with a similar customer base size to Central — investigate whether this is a coverage, pricing, or assortment gap, and pilot region-specific promotions or sales outreach there.

2. Bundle high-AOV Technology products with everyday Office Supplies.

Technology drives the most revenue per order; pairing big-ticket items (copiers, machines, phones) with complementary supplies at checkout can lift the AOV of Office Supplies orders, which currently pulls down the overall average despite having the most order volume.

3. Plan inventory and staffing around the Q4 seasonal peak.

Revenue consistently peaks in November/December. Ensure stock levels, warehouse capacity, and shipping partners are scaled ahead of this window, and consider extending seasonal promotions into the traditionally slower Q1 to smooth demand.

4. Review or discontinue chronic low-revenue SKUs.

Products like paper clips, rubber bands, and low-cost binders generate under \$15 in total revenue despite multiple orders. Evaluate whether these are loss-leaders worth keeping for basket-building, or candidates for catalog pruning.

5. Build a dedicated motion around high-ticket equipment sales.

A handful of products (copiers, binding systems, videoconferencing units) generate outsized revenue from very few orders. Enterprise/B2B outreach, financing options, or volume discounts targeting these categories could meaningfully move total revenue with relatively few additional transactions.

Data note: these recommendations are based on revenue alone. To fully validate them — especially SKU pruning and bundling decisions — Alfido Tech should incorporate Profit, Discount, and Quantity data (available in fuller Superstore extracts) so that margin, not just top-line revenue, drives the final call.